

1. Alice rolls a 6-sided dice until her dice shows a value of “3” for the second time, and then she stops. Let X denote the number of rolls needed until the second value of “3” occurs. Give an expression for $P(X > r)$, where r is any fixed positive integer value.

2. Consider two independent Geometric random variables, X and Y . Suppose that X has probability mass function $p_X(x) = P(X = x) = (3/5)^{x-1}(2/5)$ for integers $x \geq 1$. Suppose that Y has probability mass function $p_Y(y) = P(Y = y) = (4/9)^{y-1}(5/9)$ for integers $y \geq 1$.

2a. Define a new random variable $W = X + Y$. Is W a Negative Binomial random variable? Why or why not?

2b. Define another new random variable $V = 10X = X + X + \cdots + X$, i.e., V is equal to the same random variable X added to itself, ten times. Is V a Negative Binomial random variable? Why or why not?

3. Consider a 6-sided dice with 1 red side, 2 blue sides, and 3 yellow sides. Catherine rolls the dice repeatedly, until the red side appears for the 9th time, and then she stops. Let X denote the number of blue results during this process. Find the expected value of X .

4. Consider the independent random variables X and Y from Question **2**.

4a. Find $E(X + Y)$.

4b. Find $E(X - Y)$.

4c. Find $Var(X + Y)$.

4d. Find $Var(X - Y)$.

5. Let X , Y , and Z be independent Geometric random variables that each have expected value $1/p$, i.e., $p_X(x) = (1-p)^{x-1}p$ for integers $x \geq 1$, and $p_Y(y) = (1-p)^{y-1}p$ for integers $y \geq 1$, and $p_Z(z) = (1-p)^{z-1}p$ for integers $z \geq 1$.

5a. Define a new random variable $V = X + Y + Z$. Is V a Negative Binomial random variable?

5b. Find $P(V > 10)$ in terms of p .

5c. Fix *any integer value of* $k \geq 2$. Find $P(V > k)$ in terms of p and k .